Thomas Kaminsky
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EDUCATION

Harvard University Cambridge, MA
A.B. in Mathematics and Statistics, Expected May 2024
Secondary in Computer Science; GPA: 3.96

Relevant Coursework (* denotes Graduate-level course): Inverse Problems in Reinforcement Learning*, Machine Learning, Reinforcement Learning, Unsupervised Learning, Real Analysis, Linear and Generalized Linear Models*, Systems Programming, Digital Fabrication.

HONORS AND GRANTS

John Harvard Scholar, Harvard University (2020—2022)
Detur Book Prize Winner, Harvard University (2022)
NSF REU Grant, University of Washington (2022)
National Merit Scholar, National Merit Scholarship Corporation (2020)
YoungArts Winner in Classical Music–Flute, YoungArts Foundation (2019)

RESEARCH EXPERIENCE

Harvard University, Department of Statistics Cambridge, MA
Senior Thesis (Ongoing) August 2023—Present
Advisor: Professor Gabriel Kreiman.
Currently pursuing a capstone thesis concerning curriculum learning. Though Reinforcement Learning autocurricula have been used to remarkable effect in diverse domains, there remains a gap in creating useful and feasible evaluations for comparing methods in general. My thesis seeks to provide a common framework for designing and evaluating autocurricula which enables researchers to perform comprehensive experiments on general Gym environments.

University of Washington, Amazon Science Hub Seattle, WA
REU Student in Computer Vision and Robotics Summer 2022, 2023 [10 Weeks]
Coordinated with members of UW, NVIDIA, and Amazon Robotics to implement a pipeline for reasoning over bin scenes for online grasping of unknown objects in using a robotic arm. Trained and refined models for segmenting unknown object instances in bin scenes, generated novel task-specific synthetic data, and applied computer vision techniques to track objects over time.

University of Minnesota, Robotics: Perception and Manipulation Lab Minneapolis, MN
Research Intern in Robotics Feb—June 2023
Worked with Professor Karthik Desingh at the University of Minnesota on problems related to pose estimation for manipulating objects with discrete and continuous symmetries, such as bottles and caps. Created a synthetic dataset using Blender and implemented energy-based implicit distribution approximators for learning orientation uncertainty. Used ROS and AprilTags to create orientation dataset collection pipeline on our dual robotic arms.

TEACHING

Harvard University, Department of Computer Science Cambridge, MA
Introduction to Reinforcement Learning (CS 184) Sept 2023—Present
Spent approximately 12 hours per week writing and grading code problem sets, and tests, teaching section, and holding office hours for CS 184, an advanced undergraduate RL course.

Harvard University, Department of Mathematics Cambridge, MA
Vector Calculus and Linear Algebra I / II (Math 22A/B) Sept 2021—May 2023
Spend approximately 12 hours per week grading problem sets, running office hours, and tutoring students in Math 22A/B, an advanced introduction to proof, linear algebra, and vector calculus, and Math 121, an advanced second course in linear algebra.

PROJECTS

Active Inverse Reinforcement Learning for Context-Conditioned MDPs (2023)
As a cumulative term paper, my partner and I developed a framework for actively gathering expert data for learning multi-task rewards and policies for a custom Minigrid environment. We implemented models for behavior cloning and network-based IRL, and evaluated our results using a number of active learning heuristics for randomized mazes and goals.

Evaluating (*k-l*)-Means Clustering of Trajectory Data (2023)
Wrote a final paper on clustering trajectories using the (k,l)-means clustering algorithm. I collected a GPS dataset of walks around Harvard Yard and clustered the data using Python, and prepared synthetic trajectory data to assess performance across changes in trajectory structure.

Assessing Performance Degradation in Copyright-Protected Diffusion Models (2023)
As a final project for my statistics junior tutorial, I considered the impact of a new framework for copyright protecting diffusion models. I reimplemented their CP-k algorithm from pseudocode and conducted experiments comparing model performance degradation and copyright safety, as well as compute time, on a diffusion model trained on color images from the CIFAR-10 dataset.

Now You See Me: Segmenting Unknown Object Instances For Warehouse Grasping (2022)
I applied recent methods from Fox lab to detect and track unknown object instances over time in warehouse bin scenes. The system was evaluated by Amazon in August 2022.. Models were trained using PyTorch, data was generated with NViSII, a Python- based renderer from NVIDIA, and the perception was integrated with the robot and camera using ROS.

Comparing Feature Representations for Adversarial Robustness (2022)
Wrote a final project assessing the impact of feature transformations on adversarial robustness against an untargeted projected gradient descent attack. Models implemented using PyTorch, and attacks using FoolBox.

SKILLS

Programming: Python, Pytorch, Tensorflow, Weights & Biases, Numpy, CUDA, Gym, MuJoCo, ROS, R, Java, C, C++, Git.

Software: LaTeX, Blender, Linux, Windows OS, AWS, SLURM.